Grade: REVIEW_REQUIRED Check Plus

Assignment 1 (Dhaval Patel / dp129@stern.nyu.edu)

Reviewer Notes:
x Qa: Review manually.
x Qb: Review manually.
x Qc: Review manually.
x Qd: Review manually.

(1) What is the probability that a randomly selected customer withdraws at least $60 from the ATM?

P(w ≥ 60) = f(60) + f(80) + f(100)
= 0.15 + 0.25 + 0.05
= 0.45 or 45%

(2) Show by calculation that the expected value withdrawn from the ATM is $50 and that the standard deviation of the amount withdrawn from this ATM is $26.27.

x	f(x)	xf(x)	$(x-50)^2$ f(x)
10	0.10	1.00	160
20	0.15	3.00	135
40	0.30	12.00	30
60	0.15	9.00	15
80	0.25	20.00	225
100	0.05	5.00	125
	1.00	50.00	690

The Expected Value $\mu = \sum_{all\ x} xf(x) = 1 + 3 + 12 + 9 + 20 + 5 = 50$

The Standard Deviation $\sigma = \sqrt{\sum_{all\ x} (x - \mu)^2 f(x)}$

$= \sqrt{\sum_{all\ x} (x - 50)^2 f(x)}$

$= \sqrt{160 + 135 + 30 + 15 + 225 + 125}$

$= \sqrt{690}$

$= 26.27$

(3) The manager of this convenience score tells you that an expected value of $50 is not correct because it is impossible to withdraw $50 from the ATM. How would you answer the manager?

x Qe: Review manually.

The expected value of $50 is represented over the long run and can be known as the long-run average. While $50 cannot be withdrawn physically, the amount is the mathematical average of all withdrawals over a given period of time.

(4) What is the probability that the amount withdrawn from the ATM will fall within one standard deviation of the mean?

$P(50.00 - 26.27 \leq w \leq 50.00 + 26.27) =$
$P(23.73 \leq w \leq 76.27) = f(40) + f(60)$
$= 0.30 + 0.15$
$= 0.45$ or 45%

(5) What is the probability that the amount withdrawn from the ATM will fall within two standard deviations of the mean?

$P(50.00 - 2*26.27 \leq w \leq 50.00 + 2*26.27) =$
$P(-2.54 \leq w \leq 102.54) = f(10) + f(20) + f(40) + f(60) + f(80) + f(100)$
$= 0.10 + 0.15 + 0.30 + 0.15 + 0.25 + 0.05$
$= 100$ or 100%